Point-Source Neutrino Search: Analysis of NGC 1068 with IceCube Data

Master's Degree in Physics

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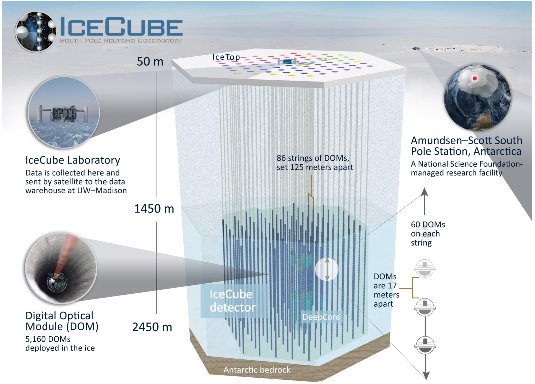
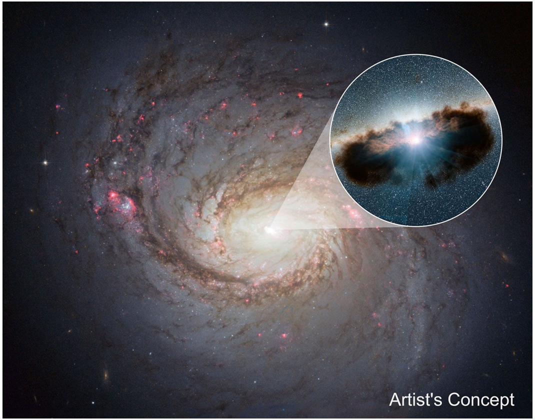
The IceCube Observatory and NGC 1068





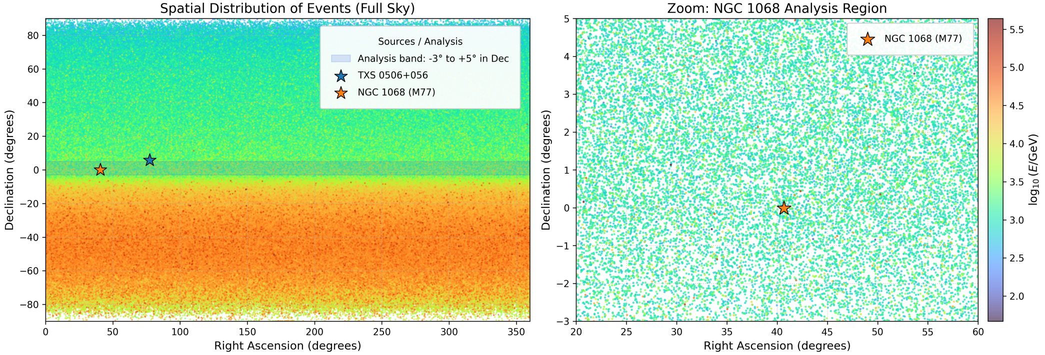
Source: icecube.wisc.edu

Source: science.nasa.gov

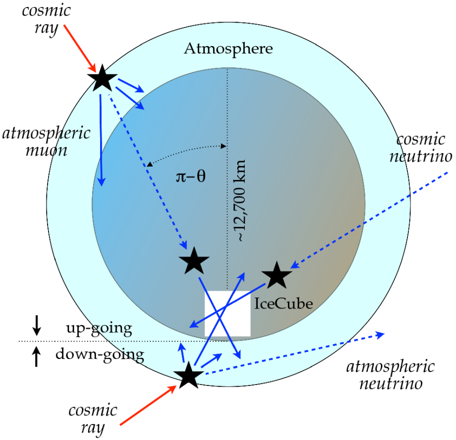
Dataset Selection & Spatial Cuts

The **Tracks dataset** provides the excellent angular resolution ($\lesssim 1^\circ$) required for point-source astronomy.

The analysis is strictly confined to a narrow **declination band** ($\delta \in [-3^\circ, +5^\circ]$).



Physical Motivation: The Declination Band



Source: link.springer.com

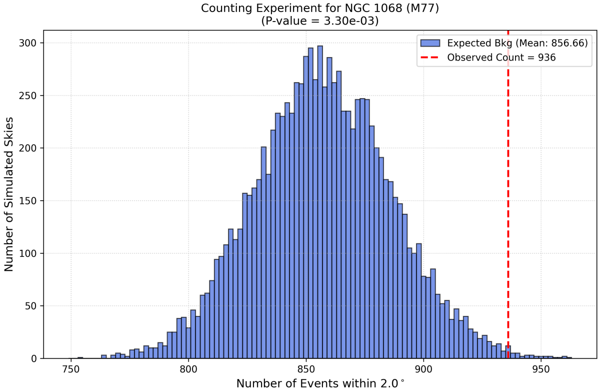
1. **Declination Band**
   - **Target:** NGC 1068 ($\delta \approx -0.06^\circ$)
   - **Window:** $\delta \in [-3^\circ, +5^\circ]$

   $\rightarrow$ Uniform detector acceptance

2. **Background Estimation**
   - **Method:** RA Scrambling
   - **Action:** Fix δ, randomize RA

   $\rightarrow$ Data-driven background (incorporates exposure)

First naive signal search for NGC 1068



Counting Analysis Results

Simulated Skies: 10^4

Observed Events: 936

Expected Bkg: 856.7 ± 29.1

Excess Events: 79.3

p-value: 3.3×10^{-3}

Significance: 2.72σ

Unbinned Spatial Likelihood

Mathematical Framework

$$\mathcal{L}(n_s) = \prod_{i=1}^{N} \left[\frac{n_s}{N} \mathcal{S}_i + \left(1 - \frac{n_s}{N}\right) \mathcal{B}_i \right]$$

- N: Total events in the declination band.
- n_s: Number of signal events (*free parameter* to be maximized).

Probability Density Functions (PDFs)

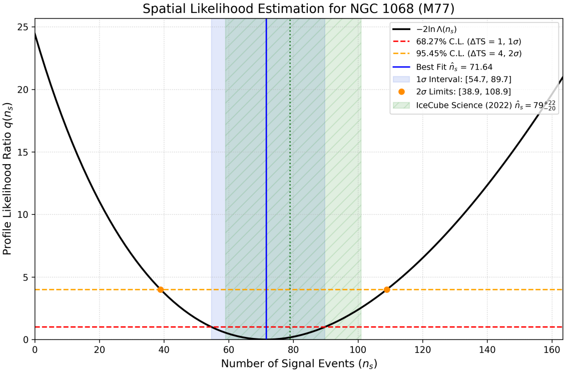
Signal PDF ($\mathcal{S}_i$)

$$\mathcal{S}_i = \frac{1}{2\pi\sigma_i^2} \exp\left(-\frac{\psi_i^2}{2\sigma_i^2}\right)$$

Background PDF ($\mathcal{B}_i$)

$$\mathcal{B}_i(\delta_i) = \frac{1}{\Delta\Omega_{\text{band}}} = \frac{1}{2\pi(\sin\delta_{\text{max}} - \sin\delta_{\text{min}})}$$

Results: Unbinned Spatial Likelihood Scan



Best-Fit Signal

$\hat{n}_s = 71.64^{+18.08}_{-16.94}$ (1σ C.L.)

2σ interval: $[38.94, 108.91]$

Statistical Significance

$TS_{\text{obs}} = 24.46$

Local p-value $< 2.0 \times 10^{-4}$

(0 background fluctuations in 5000 trials)

Significance: $\geq 3.54\sigma$

Future Directions

Enhancing the Likelihood Model:

- **Energy Integration:** Add energy-dependent PDFs (e.g., injecting an $E^{-3.2}$ source spectrum) to boost signal-background discrimination.
- **Source Morphology:** Test extended spatial models instead of a strict point-source approximation.

Astrophysical Context:

- **Time-Dependent Search:** Investigate temporal correlations with known AGN flaring activities.
- **Multi-Messenger Approach:** Cross-correlate neutrino arrivals with X-ray and γ-ray astronomical catalogs.

Conclusions

- ✓ **Methodology:** Successfully implemented a complete, unbinned spatial likelihood pipeline using a data-driven background (RA scrambling).
- ✓ **Detection:** Found a statistically significant neutrino excess ($\geq 3.54\sigma$) originating from the NGC 1068 coordinates.
- ✓ **Consistency:** Despite the "spatial-only" simplification, the estimated signal ($\hat{n}_s \approx 72$) is remarkably consistent with the complex official IceCube findings.

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Thank you for listening!

Any questions?